

1. INPUT REDIRECTION — stdin

```
saireddy@DESKTOP-HKQRL9G:~/MultiPipe_Practical$ mkdir Input_Redirection
cd Input_Redirection
saireddy@DESKTOP-HKQRL9G:~/MultiPipe_Practical/Input_Redirection$ nano input.txt
saireddy@DESKTOP-HKQRL9G:~/MultiPipe_Practical/Input_Redirection$ nano input.c
saireddy@DESKTOP-HKQRL9G:~/MultiPipe_Practical/Input_Redirection$ gcc input.c -o input
saireddy@DESKTOP-HKQRL9G:~/MultiPipe_Practical/Input_Redirection$ ./input < input.txt
Reading input from stdin:
Hello Linux
```

```
GNU nano 7.2
#include <stdio.h>

int main()
{
    char ch;

    printf("Reading input from stdin:\n");

    while ((ch = getchar()) != EOF)
    {
        putchar(ch);
    }

    return 0;
}
```

```
saireddy@DESKTOP-HKQRL9G:~/MultiPipe_Practical/Input_Redirection$ nano input.txt
saireddy@DESKTOP-HKQRL9G:~/MultiPipe_Practical/Input_Redirection$ nano input.c
saireddy@DESKTOP-HKQRL9G:~/MultiPipe_Practical/Input_Redirection$ gcc input.c -o input
saireddy@DESKTOP-HKQRL9G:~/MultiPipe_Practical/Input_Redirection$ ./input < input.txt
Reading input from stdin:
Hello Linux
This is input redirection.
saireddy@DESKTOP-HKQRL9G:~/MultiPipe_Practical/Input_Redirection$ |
```

2. OUTPUT REDIRECTION — stdout

```
This is input redirection.
saireddy@DESKTOP-HKQRL9G:~/MultiPipe_Practical/Input_Redirection$ < input.txt
saireddy@DESKTOP-HKQRL9G:~/MultiPipe_Practical/Input_Redirection$ nano output.c
saireddy@DESKTOP-HKQRL9G:~/MultiPipe_Practical/Input_Redirection$ gcc output.c -o output
saireddy@DESKTOP-HKQRL9G:~/MultiPipe_Practical/Input_Redirection$ ./output > output.txt
saireddy@DESKTOP-HKQRL9G:~/MultiPipe_Practical/Input_Redirection$ cat output.txt
Hello Linux
This output is redirected to a file.
saireddy@DESKTOP-HKQRL9G:~/MultiPipe_Practical/Input_Redirection$ |
```

3. APPEND REDIRECTION — >>

```
saireddy@DESKTOP-HKQRL9G: ~
GNU nano 7.2
#include <stdio.h>

int main()
{
    printf("New line added to the file.\n");
    return 0;
}
```

This output is redirected to a file.

```
saireddy@DESKTOP-HKQRL9G:~/MultiPipe_Practical/Input_Redirection$ > output.txt
saireddy@DESKTOP-HKQRL9G:~/MultiPipe_Practical/Input_Redirection$ nano append.c
saireddy@DESKTOP-HKQRL9G:~/MultiPipe_Practical/Input_Redirection$ gcc append.c -o append
saireddy@DESKTOP-HKQRL9G:~/MultiPipe_Practical/Input_Redirection$ ./append >> output.txt
saireddy@DESKTOP-HKQRL9G:~/MultiPipe_Practical/Input_Redirection$ cat output.txt
New line added to the file.
saireddy@DESKTOP-HKQRL9G:~/MultiPipe_Practical/Input_Redirection$ |
```

4. ERROR REDIRECTION — stderr

```
saireddy@DESKTOP-HKQRL9G: ~
GNU nano 7.2
#include <stdio.h>

int main()
{
    FILE *fp;

    fp = fopen("missing.txt", "r");

    if (fp == NULL)
    {
        fprintf(stderr, "Error: Unable to open missing.txt\n");
        return 1;
    }

    fclose(fp);

    return 0;
}
```

New line added to the file.

```
saireddy@DESKTOP-HKQRL9G:~/MultiPipe_Practical/Input_Redirection$ nano error.c
saireddy@DESKTOP-HKQRL9G:~/MultiPipe_Practical/Input_Redirection$ gcc error.c -o error
saireddy@DESKTOP-HKQRL9G:~/MultiPipe_Practical/Input_Redirection$ ./error
Error: Unable to open missing.txt
saireddy@DESKTOP-HKQRL9G:~/MultiPipe_Practical/Input_Redirection$ |
```

```
saireddy@DESKTOP-HKQRL9G:~/MultiPipe_Practical/Input_Redirection$ cat output.txt
New line added to the file.
saireddy@DESKTOP-HKQRL9G:~/MultiPipe_Practical/Input_Redirection$ nano error.c
saireddy@DESKTOP-HKQRL9G:~/MultiPipe_Practical/Input_Redirection$ gcc error.c -o error
saireddy@DESKTOP-HKQRL9G:~/MultiPipe_Practical/Input_Redirection$ ./error
Error: Unable to open missing.txt
saireddy@DESKTOP-HKQRL9G:~/MultiPipe_Practical/Input_Redirection$ ./error 2> error.txt
saireddy@DESKTOP-HKQRL9G:~/MultiPipe_Practical/Input_Redirection$ cat error.txt
Error: Unable to open missing.txt
saireddy@DESKTOP-HKQRL9G:~/MultiPipe_Practical/Input_Redirection$ |
```